

OPJ Girls College F-8/2, Islamabad

O.P.F Girls College Islamabad



Signature Of Invigilator

[Signature]

Copy No:

Name: Aleemab Alif

Roll No. 24683

Class: M-II

Section: H

Subject: Chemistry

Date: 24-August-2020

No Of Continuation Sheets Attached

Question No.	1	2	3	4	5	6	7	8	9	10	11	12	Total	Out of
Marks Obtained	8	24	10	5									43	50

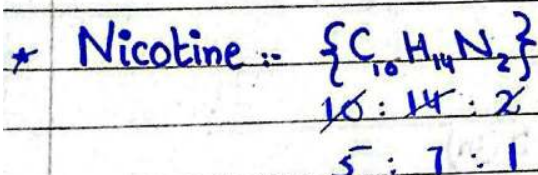
Start writing here

SECTION - C₃B

Q1

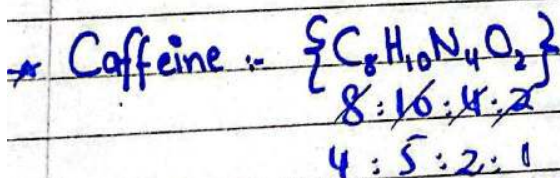
Question 1 (or)

Answer 1 (or)



Empirical formula: C_5H_7N ✓

3



Empirical formula: $C_4H_5N_2O$ ✓

Question 2 (or)

Answer 2 (or)

Effect of temperature on solubility :-

* Increase of temperature on solubility. E.g KCl

* Decrease of temperature on solubility. E.g NaCl

3

Question 3

Answer 3

An atom is electrically neutral.

An atom is electrically neutral because the total number of protons is equal to the total numbers of neutron electrons in the electron cloud.

Question 4 (or)

Answer 4 (or)

Shielding effect decreasing :-
Increasing ionization energy.

$Li < Na < K$

Iodine < Bromine < Chlorine

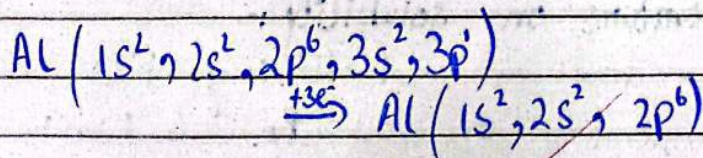
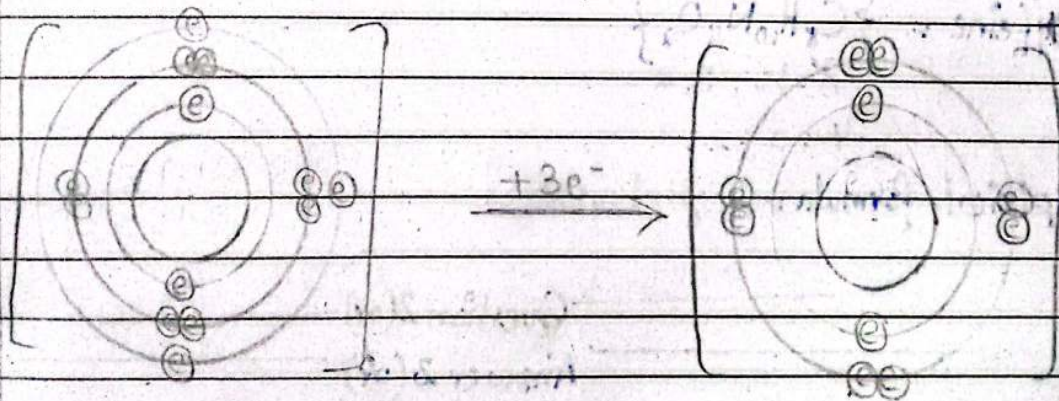
reason:-

When you move across the period, shielding effect energy increases remains constant but at some time it starts to decrease.

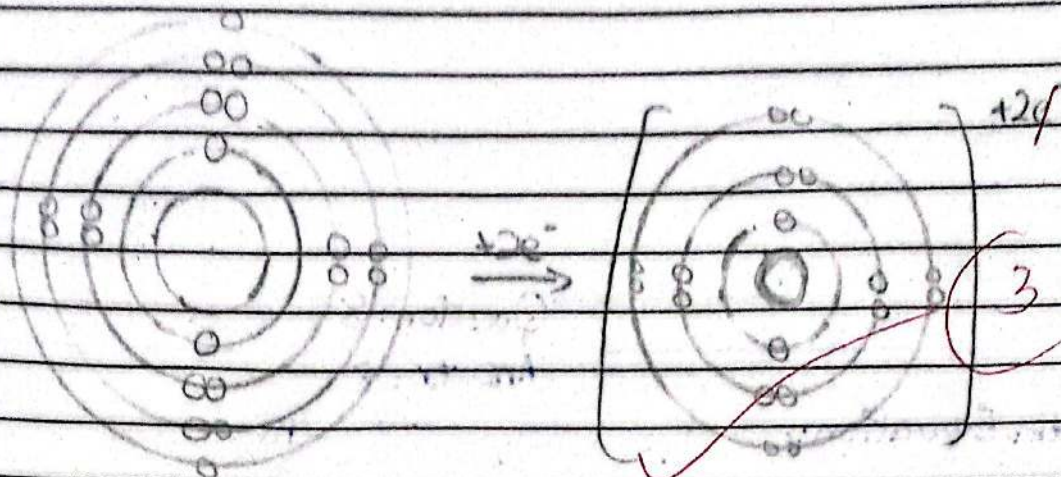
Question: 5 (or)

Answer: 5 (or)

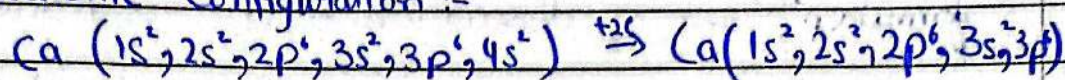
@ $Al(2:13)$



b) Ca (2, 20)



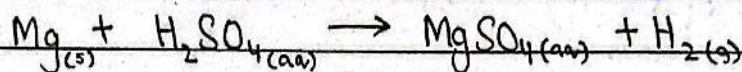
Electronic Configuration :-



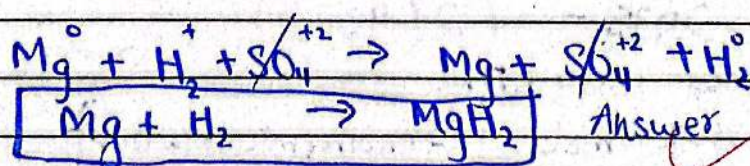
Question: 6

Answer: 6

Equation:-



Converting into ionic equation:-



Question 7 (or)

Answer: 7 (or)

a) N in HNO_3

$$+1 + x + 3(-2) = 0$$

$$1 + x - 6 = 0$$

$$x - 6 + 1 = 0$$

$$x - 5 = 0$$

$$x = +5$$

⑥ N in NO_2

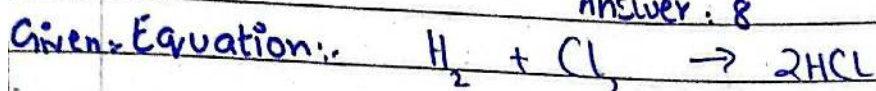
$$x + 2(-2) = 0$$

$$x - 4 = 0$$

$$x = +4$$

Question: 8

Answer: 8



formula: $\Delta H^\circ = \text{Sum of dissociation energy for reactants}$

$-\text{Sum of formation bonds for products}$

Calculation:

$$\Delta H^\circ = [(\text{BDE of H-H}) + (\text{BDE of Cl-Cl})] - (\text{BFE of } 2 \text{ H-Cl})$$

$$\Delta H^\circ = (436 \text{ kJ/mol} + 243 \text{ kJ/mol}) - (2 \times 432 \text{ kJ/mol})$$

$$= 679 \text{ kJ/mol} - 864 \text{ kJ/mol}$$

$$\Delta H^\circ = -185 \text{ kJ/mol} \quad \{\text{exothermic reaction}\}$$

Question: 9

Answer: 9

Reversible reaction

Forward reaction

Definition

① Reverse reactions are those reactions which have both direction simultaneously.

Forward reaction are those reactions which have only one direction which is forward.

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 22/8

Copy No:

Name: Seemab Atif

Roll No. 24683

Class: M-II Section: H Subject: Chemistry

Date: 24-Aug-2026 No Of Continuation Sheets Attached

Question No.	1	2	3	4	5	6	7	8	9	10	11	12	Total	Out of
Marks Obtained														

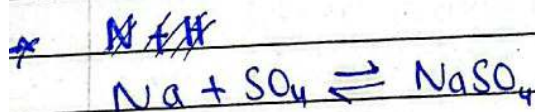
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Representation

* It is represented by double arrows ($\rightleftharpoons$)

It is represented by single arrow ($\rightarrow$)

Example

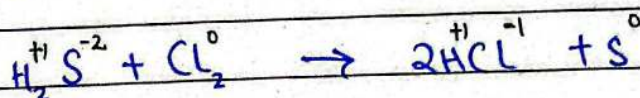
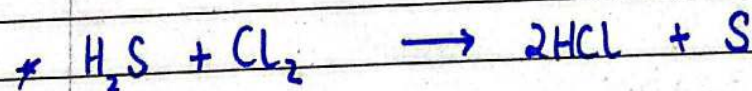


3

SECTION - C.

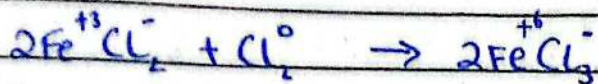
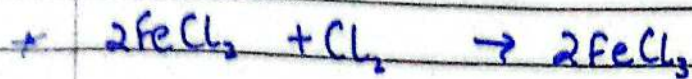
Question: 1

Answer: 1



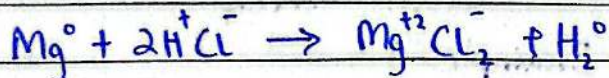
① H_2S {S} is oxidized (from -2 to 0) {Reducing agent} because it loses hydrogen

② Cl_2 is reduced (from 0 to -1) {oxidizing agent} because it gains hydrogen.



① 2FeCl_2 { Fe } is oxidized (from +3 to +6)
 { reducing agent }

② Cl_2 is reduced { 0 to -1 } { oxidizing agent }



Mg is oxidized { 0 to +2 } (reducing agent)
 2HCl { H } is reduced { +1 to 0 } (oxidizing agent)

Question : 2 (or)

Answer : 2 (or)

G. mass : ?

molecules : 3×10^{10}

molar mass
 moles of O_2 : 2 (16)

32 g/mol

formula :

molecules : $\text{mass} \times N_A$

mass : $\frac{\text{molecules}}{N_A}$

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Copy No:

Name: Ademab Afif

Class: M-II

Date: 24-Aug-2026

Section: 18

Subject: Chemistry

Roll No: 24683

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Bguf

Question No.	1	2	3	4	5	6	7	8	9	10	11	12	Total	Out of
Marks Obtained														

Start writing here

$$\text{mass} = \frac{3 \times 10^{10}}{6.022 \times 10^{23}}$$

$$\text{mass} = \left[\frac{3}{6.022} \right] 10^{10} \times 10^{-23}$$

$$\text{mass} = 0.498 \times 10^{-13}$$

$$\text{mass} = 4.98 \times 10^{-14}$$

$$\text{mass} = 4.98 \times 10^{-14} \text{ g/mol}$$

4.5